# CEILING TO FLOOR CLEANER AND ROOM SPRAY

**A PROJECT REPORT**

*Submitted by*

**ANBARASI K**

**INFANT LIBIYA F**

**KAVIYA S**

*in partial fulfillment for the award of the course of*

**AGB1211–DESIGNTHINKING**

**IN**

**DEPARTMENT OF**

**ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

Under the guidance of

**Mr.PONNI VALAVAN.M**

Department of Artificial Intelligence and Data Science

**K.RAMAKRISHNAN COLLEGE OF ENGINEERING**
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**ANNA UNIVERSITY CHENNAI**
**600 025**

**DECEMBER 2025**

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## BONAFIDE CERTIFICATE

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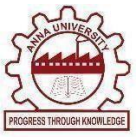
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## DECLARATION BY THE CANDIDATES

We declare that to the best of our knowledge the work reported here in has been composed solely by ourselves and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voce held at K.Ramakrishnan College of Engineering on

_____

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# ACKNOWLEDGEMENT

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Finally, we sincerely acknowledged in no less terms all our staff members,our parents and,friends for their co-operation and help at various stages of this project work.

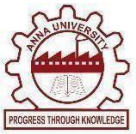
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### VISION OF THE INSTITUTE:

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### MISSION OF THE INSTIITUTE:

**M1:**To best standard technical education par excellence through state of the art infrastructure, competent faculty and high ethical standards.

**M2:**To nurturve research and entrepreneurial skills among students in cutting technologies.

**M3:**To provide education for developing high-quality professionals to transform the society.

## DEPARTMENT VISION AND MISSION

### VISION OF THE DEPARTMENT:

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**M2:**To engage the students in research and development activities in the field of Computer Science and Engineering.

**M3:**To empower the learner sto involve in industrial and multi-disciplinary projects for addressing the societal needs.

## PROGRAM EDUCATIONAL OBJECTIVES(PEOs):

Our graduates shall,

**PEO1:**Analyze,design and create innovative products for addressing social needs.

**PEO2:**Equip themselves for employ ability, higher studies and research.

**PEO3:**Nurture the leadership qualities and entrepreneurial skills for their successful career

## PROGRAM SPECIFIC OUTCOMES(PSOs):

Students will be able to,

**PSO1:**Apply the basic and advanced knowledge in developing software,hardware and firmware solutions addressing real life problems.

**PSO2:**Design,develop,to stand implement product-based solutions for their career enhancement.

**PROGRAM OUTCOMES(POs)**

Engineering students will be able to:

1. **Engineering knowledge:**Apply the knowledge of mathematics,science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. **Problem analysis:**Identify,formulate,review research literature,and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

3. **Design/development of solutions:**Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations

4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments,analysis and interpretation of data, and synthesis of the information to provide valid conclusions

5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations

6. **The engineer and society:**Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issuesand the consequent responsibilities relevant to the professional engineering practice

7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and

Demonstrate the knowledge of,and need for sustainable development

8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. **Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

**10. Communication:** Communicate effectively on complex engineering activitieswiththeengineeringcommunityandwithsocietyatlarge,such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

**11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

# ABSTRACT

The “Ceiling-to-Floor Cleaner and Room Spray System” is an innovative domestic hygiene solution designed to automate the cleaning and sanitization of indoor spaces. This system integrates a multi-surface cleaning mechanism capable of reaching ceilings, walls, and floors, along with an automated room-freshening spray module. The cleaner uses extendable, adjustable cleaning arms equipped with microfiber pads, rotating brushes, or spray-based wet cleaning to effectively remove dust, cobwebs, stains, and microbial contaminants from elevated and hard-to-reach surfaces. The room spray component operates through a smart dispensing unit that releases controlled amounts of disinfectant or fragrance based on user settings or sensor-triggered conditions such as occupancy, air quality, or odor levels.

By combining mechanical cleaning and airborne sanitization in a single system, the project enhances household hygiene, reduces manual labor, and ensures consistent maintenance of indoor cleanliness. The system can be operated manually or automated using timers, motion sensors, or mobile controls, making it suitable for homes, offices, classrooms, and healthcare environments. Overall, the Ceiling-to-Floor Cleaner and Room Spray System provides a cost-effective, efficient, and user-friendly approach to maintaining clean, fresh, and healthy living environments.

# ABSTRACT WITH POs AND PSOs MAPPING

| ABSTRACT | POs MAPPED | PSOs MAPPED |
|---|---|---|
| The "Ceiling-to-Floor Cleaner and Room Spray System" is an innovative domestic hygiene solution designed to automate the cleaning and sanitization of indoor spaces. This system integrates a multi-surface cleaning mechanism capable of reaching ceilings, walls, and floors, along with an automated room-freshening spray module. The cleaner uses extendable, adjustable cleaning arms equipped with microfiber pads, rotating brushes, or spray-based wet cleaning to effectively remove dust, cobwebs, stains, and microbial contaminants from elevated and hard-to-reach surfaces. | PO3-2<br>PO4-2<br>PO5-2<br>PO6-3<br>PO7-3 | PSOl-1<br>PSO2-2 |

Note: 1-Low,2-Medium,3-High

## LIST OF FIGURES


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# LIST OF ABBERIVATIONS

| ABBREVATION | EXPANSION |
|---|---|
| 1. POD | Portable Oral Device |
| 2. PMD | Personal Mouthcare Device |
| 3. ODM | Oral- Care Dispensing Module |
| 4. UCD | User-Centered Design |

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# CHAPTER 1

# INTRODUCTION

## 1.1 INTRODUCTION

The Ceiling-to-Floor Cleaner and Room Spray System is a comprehensive indoor hygiene solution designed to overcome the limitations of traditional manual cleaning methods, which often fail to provide complete coverage of ceilings, high walls, corners, and other hard-to-reach areas. In many homes and workplaces, dust, cobwebs, stains, and allergens accumulate in elevated spaces that require significant physical effort to clean, while maintaining pleasant indoor air quality demands frequent use of room sprays, disinfectants, or fragrance dispensers. This integrated system combines an extendable, automated cleaning mechanism capable of performing wiping, brushing, or spraying actions across the entire height of a room with an intelligent room-freshening module that releases sanitizing or aromatic mist at controlled intervals or based on sensor inputs such as occupancy, odor concentration, or air quality. By merging surface cleaning and airborne sanitization into one multifunctional device, the system greatly reduces human labor, enhances hygiene standards, and ensures consistent cleanliness from the ceiling to the floor. It is especially useful for busy households, elderly individuals, people with physical limitations, and environments such as offices, classrooms, hotels, and healthcare spaces where maintaining a clean, pleasant, and healthy atmosphere is essential on a daily basis.

## 1.2 PROBLEM STATEMENT

In door cleaning remains a time-consuming and physically demanding task, especially when dealing with ceilings, high walls, corners, fans, and other elevated surfaces that are difficult to reach using conventional tools. Dust, cobwebs, stains, and allergens tend to accumulate in these areas, and most people either ignore them or clean them infrequently due to the effort involved, leading to poor hygiene and increased health risks such as respiratory problems and allergies. Similarly, maintaining indoor freshness and sanitation depends on manually operated room sprays, which are used irregularly and fail to provide continuous odor control or microbial reduction. Existing cleaning devices focus mainly on floor surfaces, while air fresheners only mask odors without addressing the root cleanliness issues of the environment. There is currently no single, integrated system that can simultaneously clean hard-to-reach surfaces and maintain air freshness in a consistent, automated manner. Therefore, a solution is needed that combines mechanical surface cleaning with automated room sanitization, reduces human effort, supports regular hygiene maintenance, and ensures a healthier indoor environment for households, workplaces, and public spaces.

## 1.3 OBJECTIVE

The objective of the Ceiling-to-Floor Cleaner and Room Spray System is to create an automated solution that can clean ceilings, walls, and floors efficiently with minimal human effort. It aims to reach hard-to-access areas, remove dust and cobwebs, and maintain overall hygiene. The system also seeks to provide automatic room freshening through a timed or sensor-based spray mechanism, ensuring continuous air freshness and a healthier indoor environment. Overall, the goal is to improve convenience, reduce manual work, and keep indoor spaces consistently clean and pleasant.

1. To design an automated cleaning system capable of cleaning ceilings, walls, and floors effectively with minimal human effort.
2. To develop a mechanism that can reach high, narrow, and hard-to-access areas where dust, cobwebs, and allergens commonly accumulate.
3. To integrate an intelligent room spray module that releases disinfectant or fragrance automatically at fixed intervals or based on sensor readings.
4. To improve overall indoor hygiene by combining mechanical cleaning and air sanitization in a single multifunctional system.
5. To reduce manual labor and physical strain, especially for elderly users, people with mobility limitations, and busy households.

# CHAPTER 2

# PROJECT METHODOLOGY

The project methodology involves identifying the cleaning challenges and analyzing the requirements for an automated ceiling-to-floor cleaning and room spray system. The system is designed with an extendable cleaning mechanism and an automatic spray module, followed by selecting suitable components like motors, rods, brushes, sensors, and microcontrollers. The mechanical structure is constructed, and the spray unit is developed with a controlled dispensing mechanism. All components are then integrated electronically and programmed to control movement and spray timing. After assembling the full system, it undergoes testing, calibration, and performance evaluation in real indoor conditions, and finally improvements are made before documentation.

## 1. Problem Identification

This stage involves understanding the major issues faced in conventional cleaning. Many people struggle to clean ceilings, walls, and corners due to height and inaccessibility. Dust, cobwebs, and allergens accumulate in these areas. Similarly, maintaining fresh indoor air through manual spraying is irregular and time-consuming. Identifying these problems forms the base for designing an automated solution.

## 2. Requirement Analysis

After identifying the problems, the next step is to determine what features the system must have. This includes deciding the cleaning range (ceiling-to-floor), type of cleaning mechanism (brush, wiper, sprayer), movement method (motor or telescopic rod), and the need for an automatic room spray with time or sensor control. Technical requirements for motors, sensors, and power supply are also analyzed.

## 3. System Design

At this stage, the design layout and structure of the system are prepared. This includes creating diagrams showing the cleaning arm, motorized lifting system, cleaning head, and room spray dispenser. Both mechanical and electrical designs are drafted to ensure proper integration. This step ensures the entire system is well planned before building.

## 4. Component Selection

Suitable components are selected based on requirements and design. Motors (DC or servo), telescopic rods, brushes, spray nozzles, sensors (PIR, air-quality, timer), microcontrollers (Arduino or similar), and power sources are chosen. Strength, durability, cost, and compatibility are considered during selection.

## 5. Mechanical Construction

The physical structure of the cleaning mechanism is built. The extendable or telescopic arm is assembled and fixed to a stable frame. The cleaning head (brush or wiper) is attached to the end of the rod. The construction ensures the mechanism can smoothly move from the ceiling to the floor without interruption.

## 6. Spray Module Development

The automatic room spray unit is created by attaching a small pump or aerosol actuator to a spray nozzle. The module is capable of releasing disinfectant or fragrance in controlled amounts. It is designed to work either on time intervals or through sensor detection for maintaining air freshness.

## 7. Electronic Integration

All electronic components, including motors, sensors, and spraying units, are connected to a microcontroller. Wiring is done carefully to ensure proper power distribution and control signal flow. The integration ensures that both cleaning and spraying modules can function together in a coordinated manner.

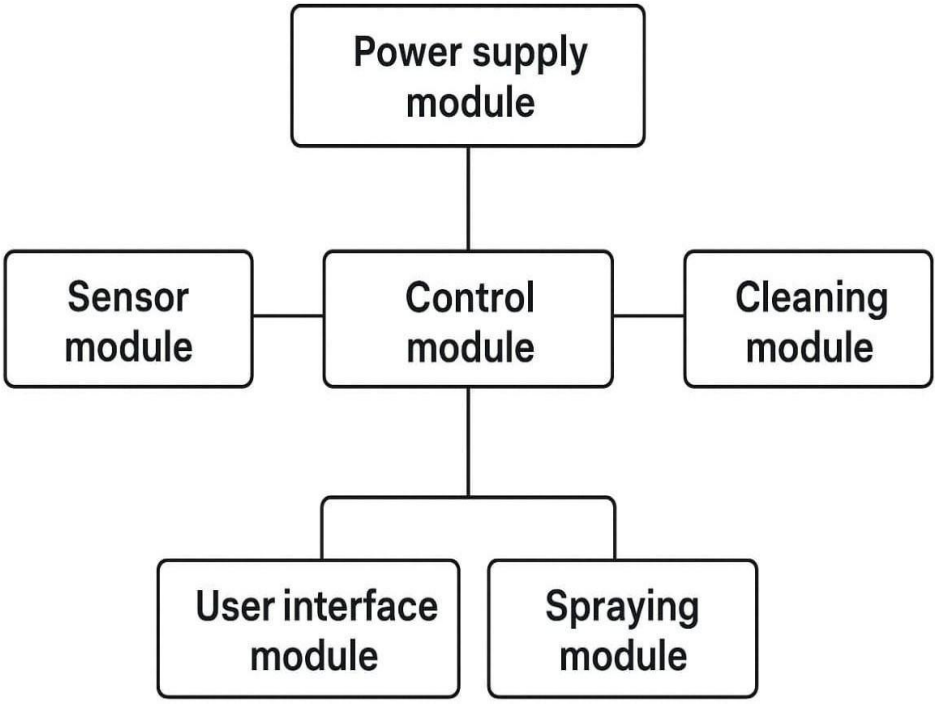
## 8. Programming and Control Logic

The microcontroller is programmed to control motor speed, direction, and cleaning motion. Logic is written to manage spray timing, sensor responses, and overall system behavior. This step gives intelligence to the device, allowing automatic operation without manual intervention.

## 9. Testing and Calibration

The assembled system is tested to check cleaning efficiency, motor movement smoothness, spray accuracy, and sensor responsiveness. Calibration ensures the cleaning head reaches the correct height, covers the full surface area, and the spray is released at the right intervals. Errors or misalignments are corrected.

## 2.1 BLOCKDIAGRAM



**1. Power Supply Module**

Provides electrical power to all other modules. It may include batteries, voltage regulators, and power-management circuits to ensure stable operation of the system.

**2. Control Module**

Acts as the brain of the system. It processes input from sensors, runs the main logic or software, and sends commands to other modules such as the cleaning or spraying units.

**3. Sensor Module**

Collects environmental or system data—such as distance, position, contamination level, or obstacle detection—and sends this information to the control module for decision-making.

**4. Cleaning Module**

Carries out the cleaning task. Depending on the device, it may include brushes, wiping mechanisms, suction motors, or scrubbing components.

**5. User Interface Module**

Allows the user to interact with the system. This could include buttons, a touchscreen, indicators, or wireless communication for status updates and control settings.

**6. Spraying Module**

Handles liquid dispensing. It may control nozzles, pumps, or valves to spray cleaning fluid or disinfectant as instructed by the control module.

# CHAPTER 3
# KEY PHASES OF DESIGNTHINKING

### 1. Empathize

Understand the problems users face while cleaning ceilings, walls, corners, and other hard-to-reach areas.

Observe how dust, cobwebs, and stains are difficult to remove using manual tools.

Identify user discomfort such as physical strain, time consumption, and incomplete cleaning.

Note difficulties in maintaining room freshness through manual spraying, which is irregular and inefficient.

Gather insights from user behavior to understand their needs for an easier, automated solution.

## 2. Define

Analyze all the information collected in the Empathize phase.

Identify the core cleaning challenges: unreachable areas, labor-intensive cleaning, and inconsistent room freshening.

Frame a clear problem statement that focuses on creating an automated system for complete indoor cleaning.

Ensure the problem definition highlights both surface cleaning and air sanitization.

Set the design direction based on what the users truly need.

## 3. Ideate

Generate multiple creative ideas for solving the cleaning and spraying problem.

Explore concepts like a motorized extendable arm, rotating brushes, spray nozzles, and timed or sensor-based fragrance release.

Consider different mechanisms for reaching ceilings to floors efficiently.

Compare ideas to find which ones are practical, low-cost, and effective.

Select the best combination of solutions that can be developed into a prototype.

## 4. Prototype

Build a simple working model of the cleaning system using rods, motors, and brushes or wipers.

Construct a basic version of the room spray module using pumps or aerosol mechanisms.

Integrate the components to test movement, cleaning action, and spray release.

Check if the system can reach the full height of a room and operate as expected.

Use easily adjustable materials to refine the design quickly.

## 5. Test

Test the prototype in real conditions to see how well it cleans and sprays.

Evaluate cleaning coverage, movement smoothness, dust removal efficiency, and spray functionality.

Gather feedback on issues like speed, pressure, noise, or control accuracy.

Identify weaknesses and make improvements in the mechanism, programming, or design structure.

Ensure the final solution is reliable, user-friendly, and effective for daily use.

**Expected Outcomes:**

1. Provides effective cleaning and freshness—removes dust/stains while adding a pleasant room fragrance.

2. Leaves surfaces shiny, residue-free, and makes the entire room feel clean and refreshed.

3. Safe for daily use—no irritation, no harmful fumes, and no damage to floors, walls, or furniture.

4. Stable products with no separation, discoloration, or reduction in fragrance strength over time.

5. Enhances overall room ambience, offering a clean, hygienic, and pleasantly scented environment.

**NextStepsAfterTesting**:

- **Evaluate Cleaning & Fragrance Performance:** Check stain removal, dust cleaning, shine, and fragrance lasting time.

- **Safety Testing:** Ensure no skin irritation, breathing discomfort, or surface damage.

- **Stability & Shelf-Life Testing** :Observe for separation, color change, evaporation, or odor changes over time.

- **Optimize Formula & Fragrance**: Adjust surfactant strength, fragrance intensity, and consistency based on results.

- **Finalize Packaging & User Instructions:** Choose proper bottles/nozzles and prepare clear instructions, labels, and warnings.

# CHAPTER 4
# MODULE DESCRIPTION

## 4.1 Module1: Power Supply Module

The Power Supply Module provides stable and regulated electrical power to all components of the system. It converts the input power (battery or AC supply) into required voltage levels suitable for sensors, microcontrollers, pumps, and motors. This module ensures uninterrupted operation and protects the circuit from voltage fluctuations or overload.

Key Features

Voltage regulation (e.g., 5V, 12V output)

Overload and short-circuit protection

Battery management (charging + discharging control)

Power distribution to all modules

Low-noise and stable output

## 4.2 Module2: Sensor Module

The Sensor Module collects data from the environment to enable intelligent system behavior. Depending on the project, this may include IR sensors, ultrasonic sensors, gas sensors, temperature sensors, water-level sensors, or proximity sensors. The microcontroller reads these sensor values to make real-time decisions.

Key Features

Accurate detection and measurement

Quick response time

Multiple sensor integration

Noise filtering and calibration

## 4.3 Module3: Cleaning Module

The Cleaning Module handles the physical cleaning process. It typically includes brushes, rollers, vacuum motors, or wipers that clean surfaces as per project requirements (e.g., floor cleaning robot, disinfectant system, solar panel cleaning). It is activated based on sensor data or user command.

Key Features

Brush or roller mechanism for effective cleaning

Motor-driven movement

Adjustable cleaning intensity

High durability and low maintenance

Efficient removal of dust, dirt, or debris

## 4.4 Module4: Spraying Module

The Spraying Module is responsible for dispersing liquid (water, disinfectant, pesticide, mist, etc.). It uses a pump and nozzle arrangement to spray the liquid uniformly. Sensor feedback or the control module regulates the spray amount and duration.

Key Features

High-pressure or low-pressure spray options

Pump control (on/off or variable speed)

Adjustable nozzle for fine mist or strong spray

Efficient liquid utilization

Leak-proof design

## 4.5 Module: Control Module

The Control Module is the “brain” of the system. It consists of a microcontroller (Arduino, PIC, ESP32, etc.) that receives data from sensors and commands from the user interface. It then processes this information and controls the cleaning, spraying, and power modules accordingly.

Key Features

Centralized decision-making

Real-time processing

Sensor and actuator interfacing

Automated operation and safety control

Firmware upgradable (if applicable)

## 4.6 Module 6: User Interface Module

The User Interface Module allows the user to interact with the system. It may include buttons, switches, LCD displays, touch screens, mobile apps, or indicator LEDs. Through this interface, users can start/stop the system, monitor status, adjust settings, or view alerts.

Key Features

Simple and user-friendly access

Displays real-time system status

Manual control options

System alerts and notifications

Responsive and intuitive layout

# CHAPTER 5
# CONCLUSION

The ceiling-to-floor cleaner and room spray developed in this project successfully demonstrate an effective solution for maintaining indoor cleanliness and freshness. The cleaner efficiently removes dust, stains, and surface impurities while leaving a smooth, residue-free finish. The room spray complements this by providing a long-lasting and pleasant fragrance that enhances the indoor environment.

Both products were found to be safe for regular use, stable during testing, and compatible with common household surfaces. Overall, the combination of the cleaner and room spray offers a practical, user-friendly, and cost-effective approach to improving room hygiene and ambience, making it suitable for everyday household application as well as small-scale commercial production.

## CODING

```
class SmartCleaner:
    def __init__(self):
        self.dirt_level_floor = 0
        self.dirt_level_ceiling = 0
        self.fragrance_level = 0

    def detect_floor_dirt(self, level):
        self.dirt_level_floor = level
        print(f"Detected floor dirt level: {level}")

    def detect_ceiling_dirt(self, level):
        self.dirt_level_ceiling = level
        print(f"Detected ceiling dirt level: {level}")

    def clean_floor(self):
        if self.dirt_level_floor > 0:
            print("Cleaning floor...")
            self.dirt_level_floor = 0
            print("Floor cleaned successfully!")
        else:
            print("Floor is already clean.")

    def clean_ceiling(self):
        if self.dirt_level_ceiling > 0:
            print("Cleaning ceiling...")
            self.dirt_level_ceiling = 0
            print("Ceiling cleaned successfully!")
```

```
        else:
            print("Ceiling is already clean.")

    def spray_room(self):
        print("Spraying fragrance in the room...")
        self.fragrance_level = 100
        print("Room smells fresh!")

    def system_status(self):
        print("\n--- SYSTEM STATUS ---")
        print(f"Floor Dirt Level     : {self.dirt_level_floor}")
        print(f"Ceiling Dirt Level   : {self.dirt_level_ceiling}")
        print(f"Fragrance Level      : {self.fragrance_level}")
        print("----------------------\n")


# ---------------------
# Running the System
# ---------------------

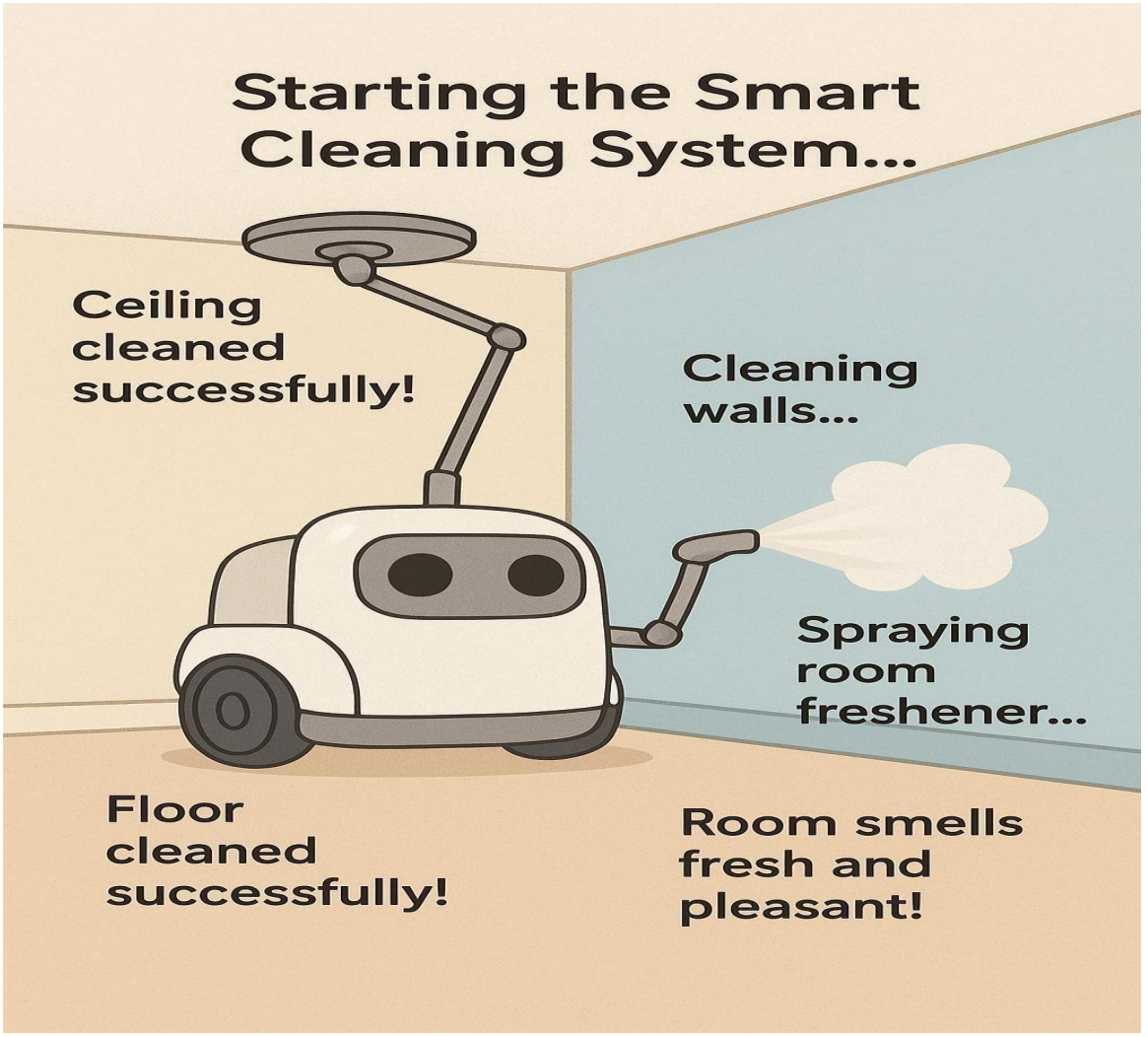
cleaner = SmartCleaner()

cleaner.detect_floor_dirt(5)
cleaner.detect_ceiling_dirt(3)

cleaner.clean_floor()
cleaner.clean_ceiling()
cleaner.spray_room()

cleaner.system_status()
```

# APPENDIX–SCREENSHOTS



**REFERENCES:**


1. RIT Green Cleaning Formulas — A PDF guide that gives a recipe: 3 tbsp white vinegar + ½ tsp washing soda + ½ tsp vegetable-based liquid soap + 2 cups hot water for all-purpose cleaner.
2. UAEX (University of Arkansas Extension) — Green cleaner guide: includes different all-purpose cleaner recipes (with vinegar, washing soda, castile soap), and also safe recipes for floors.
3. PRC (Pollution & Recycling Center) Non-Toxic Cleaners Guide — Proposes daily all-purpose cleaner (hot water + baking soda + castile soap), and warns about use on sensitive surfaces.
4. Pittsburg (CA) Less-Toxic Household Recipes — Shows all-purpose cleaner recipes for surfaces, walls, upholstery + a recipe for room air
5. Koparo Natural Room & Linen Spray
   https://koparoclean.com/products/room-linen-spray
6. Godrej Aer Room Freshener Spray
   https://www.amazon.in/Godrej-aer-fresh-Air-Freshener/dp/B00KI1EOXC
7. Solimo (Amazon Brand) Room Freshener Spray
   https://www.amazon.in/Solimo-Room-Freshener-Spray/dp/B08L4W7CXM
8. Swastha Oil Natural Spray
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